CYRUSHQ

RESEARCH GUIDE

Running Your AI CEO on a VPS

Everything you need to understand, evaluate, and set up your OpenClaw AI agent on a Virtual Private Server

Important: CyrusHQ's own AI CEO and agent team run on Mac mini computers. We have not personally tested an OpenClaw installation on a VPS. This guide is based on OpenClaw's official documentation and thorough independent research. We cannot guarantee your specific experience, but we are sharing what the research shows to help you make an informed decision.

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SECTION 01

What is a VPS?

VPS stands for **Virtual Private Server**. Think of it as renting a computer that lives inside a professional data center, rather than having a physical machine sitting on your desk at home or in your office.

When you build your AI CEO using the course, you install OpenClaw on a Mac mini — a real, physical computer that lives at your location, is always plugged in, and runs your AI agent from there. With a VPS, the "computer" is virtual: it's a dedicated slice of a powerful server in a data center that you rent by the month and access remotely over the internet.

Feature	Mac mini	VPS
Where it lives	Your home or office	A professional data center
How you access it	Directly — you sit at the keyboard or use it via your network	Remotely — via SSH (a secure terminal connection over the internet)
Operating system	macOS	Linux (Ubuntu or Debian recommended)
Cost model	One-time hardware purchase ($500–$800+)	Monthly subscription ($5–$25/mo depending on plan)
Always online?	Only when powered on at your location	Yes — runs 24/7 in the data center
Physical maintenance	You manage it yourself	Hardware is managed by the hosting provider

📡 **THE KEY IDEA**

Once you're connected to your VPS via SSH, you type commands into a terminal window on your local computer — and those commands run on the remote server in the data center. From that point on, the experience is similar to working in a Mac terminal.

SECTION 02

Does OpenClaw Support VPS?

Yes — officially. OpenClaw's documentation explicitly supports VPS installations and provides dedicated setup guides for multiple hosting providers. This is not a workaround or unsupported hack. It is a recognized and documented installation path.

According to OpenClaw's official documentation:

> ✅ **OFFICIAL OPENCLAW STATEMENT**
>
> A VPS is a good choice for an AI agent that needs to stay online continuously. OpenClaw can operate on compatible Linux systems, including properly configured Virtual Private Servers.

OpenClaw maintains dedicated installation guides for:

- **Hetzner** — docs.openclaw.ai/install/hetzner
- **DigitalOcean** — docs.openclaw.ai/install/digitalocean
- **Hostinger** — docs.openclaw.ai/install/hostinger

The core architecture of your AI CEO does not change. The same identity files, memory system, subagents, integrations, and automations you build in the course work the same way on a VPS. The workspace simply lives on the remote server instead of a local machine.

SECTION 03

System Requirements

Before choosing a VPS plan, make sure it meets OpenClaw's requirements. Here are the practical minimums and recommended specs based on OpenClaw's documentation and community research:

Component	Minimum	Recommended	Notes
Operating System	Ubuntu 22.04 LTS or Debian 11+	Ubuntu 24.04 LTS	Most stable and well-documented path
Architecture	x86_64 or ARM64	x86_64	Both are supported
RAM	1 GB (with swap)	4 GB+	Browser automation and heavy integrations need more
CPU	1 vCPU	2 vCPU	More vCPUs improve concurrent task performance
Storage	10 GB SSD	40 GB SSD	Logs, memory files, and backups grow over time
Node.js	Node.js 22.16+	Node.js 24 LTS	Install via NodeSource — distro defaults are often too old
Internet	Any standard VPS connection	—	Required for API calls and agent communication

⚠ IMPORTANT NOTE

Do not rely on local AI models (like Ollama) on a budget VPS — they require significant GPU resources. VPS installations work best when connected to external AI model APIs (OpenAI, Anthropic, etc.), the same way the course is set up.

SECTION 04

Recommended VPS Providers & Pricing

OpenClaw officially supports and provides guides for three primary providers. All pricing below is approximate and based on available documentation — verify current rates directly with each provider before purchasing.

Hetzner

~$4–$8 / month

BEST VALUE

European provider with excellent price-to-performance. CX22 plan (~€3.79/mo) offers 2 vCPU / 4 GB RAM. Strong OpenClaw documentation. Best for technically comfortable users.

DigitalOcean

~$6–$18 / month

MOST BEGINNER-FRIENDLY

Well-known US provider with easy-to-use interface. VPS products called "Droplets." Good documentation, clean control panel. Recommended for those new to cloud hosting.

Hostinger

~$5–$15 / month

EASIEST SETUP

Offers a 1-click managed OpenClaw deployment option — the fastest path to getting up and running. Also offers a manual VPS path for more control.

Additional Monthly Costs to Budget For

- **AI Model API usage** — same as with a Mac mini setup (OpenAI, Anthropic, etc.)
- **Automated backups** — many providers charge a small fee for snapshot backups
- **Additional storage** — if your workspace grows significantly
- **Tailscale** — free for personal use; optional but recommended for secure private access

💡 **COST COMPARISON PERSPECTIVE**

A basic VPS at $6–10/month works out to $72–$120/year. Compare that to a Mac mini at $500–$800 upfront. The VPS is more economical in the short term but adds up over time. A Mac mini may actually be more cost-effective over 2–3 years, and you own the hardware.

SECTION 05

What Your AI CEO Can Do on a VPS

The intelligence, personality, and capabilities of your AI CEO come from its configuration, its AI model, and its integrations — not from the hardware it runs on. A properly set-up VPS-based AI CEO can do everything covered in the course:

Capability	VPS Support
Custom AI identity, personality, and name	✓ Fully supported
Long-term memory and daily notes	✓ Fully supported
Specialized subagents	✓ Fully supported
Scheduled tasks and automations	✓ Fully supported
Discord, Telegram, and messaging integrations	✓ Fully supported
API integrations (Stripe, GoHighLevel, email, CRM)	✓ Fully supported
Web research and content generation	✓ Fully supported
Browser automation	⚠ Possible — requires headless browser setup; no local display
macOS-specific apps and local desktop access	✗ Not available on Linux VPS
Local AI model inference (Ollama, etc.)	✗ Not practical on budget VPS (needs GPU server)

The most significant difference is that a Linux VPS cannot run macOS-only software. If your AI CEO relies on macOS-specific capabilities, a VPS won't support those. For the vast majority of business use cases covered in the course, this is not a limitation.

SECTION 06

Installation Walkthrough

⚠ **RESEARCH-BASED GUIDANCE**

The steps below are based on OpenClaw's official documentation and independent research. CyrusHQ has not personally completed this installation on a VPS. Always refer to your provider's specific OpenClaw guide (docs.openclaw.ai/install/[provider]) as your primary reference, as steps may vary or be updated.

1

Choose and Provision Your VPS

Log into your chosen provider (Hetzner, DigitalOcean, or Hostinger) and create a new server. Select **Ubuntu 24.04 LTS** as your operating system. Choose a plan with at least 2 GB RAM and 20 GB storage. During setup, add your SSH key (see Step 2) rather than using a password — this is both more secure and easier to work with long-term.

2

Set Up SSH Access

SSH (Secure Shell) is how you control your VPS from your own computer. You'll need to generate an SSH key pair if you don't have one. On your local machine (Mac or PC), open Terminal and run:

```
# Generate an SSH key pair (run this on your LOCAL computer, not the VPS)
ssh-keygen -t ed25519 -C "your-email@example.com"

# Press Enter to accept defaults. Your key will be saved to
~/.ssh/id_ed25519

# Copy your public key to the VPS during server creation (paste into
provider's SSH key field)
cat ~/.ssh/id_ed25519.pub
```

Once your VPS is live, connect to it:

```
# Connect to your VPS (replace with your server's IP address)
ssh root@YOUR.SERVER.IP.ADDRESS

# You should see a Linux welcome prompt — you're now inside your VPS
```

3 Update the System & Install Prerequisites

Once connected, update the server's software and install essential tools:

```
# Update all system packages
sudo apt update && sudo apt upgrade -y

# Install basic tools
sudo apt install -y curl git ufw build-essential
```

4 Install Node.js 24 LTS

OpenClaw requires Node.js 24 LTS (or newer). The version included with Ubuntu by default is usually too old, so install it via NodeSource:

```
# Download and run the NodeSource setup script for Node.js 24
curl -fsSL https://deb.nodesource.com/setup_24.x | sudo -E bash -

# Install Node.js
sudo apt-get install -y nodejs

# Verify the installation — you should see v24.x.x or higher
node --version
v24.1.0
npm --version
```

5 Install OpenClaw

Run OpenClaw's official install script. This handles the installation automatically:

```
# Run the official OpenClaw installer
curl -fsSL https://openclaw.ai/install.sh | bash -s -- --no-onboard

# Verify the installation
openclaw --version

# Run a diagnostics check
openclaw doctor
```

6 Create a Non-Root User (Security Best Practice)

Running OpenClaw as root (the all-powerful server administrator) is risky. Create a dedicated user account for OpenClaw to run under:

```
# Create a new user (you can name it anything — "openclaw" is common)
adduser openclaw

# Give them sudo access
usermod -aG sudo openclaw

# Switch to that user
su - openclaw
```

7 Run OpenClaw Onboarding

Run the onboarding wizard, which walks you through connecting your AI model API, setting up channels (Discord, Telegram, etc.), and installing OpenClaw as a background service (daemon) so it restarts automatically:

```
openclaw onboard --install-daemon

# The wizard will prompt you for:
# - Your AI model provider and API key (OpenAI, Anthropic, etc.)
# - Messaging channels (Discord bot token, Telegram token, etc.)
# - Gateway configuration
# - Daemon/auto-start installation
```

8 Set Up Your AI CEO Workspace

Your workspace lives at `~/.openclaw/workspace/` on the VPS — the same structure used in the course. You'll create or upload your identity and configuration files here:

```
# Navigate to your workspace
cd ~/.openclaw/workspace

# Create files directly on the VPS using nano (a simple text editor)
nano AGENTS.md

# OR upload files from your local computer using SCP
```

```
# (run this from your LOCAL computer terminal)
scp ~/Desktop/AGENTS.md openclaw@YOUR.SERVER.IP:~/.openclaw/workspace/
```

The same Markdown files from the course (AGENTS.md, SOUL.md, IDENTITY.md, MEMORY.md, etc.) work exactly the same way on the VPS. This is one of the biggest advantages — your AI CEO's brain doesn't change, only where it lives.

9 Verify the Gateway is Running

Check that OpenClaw is running as a background service:

```
# Check gateway status
openclaw gateway status

# View live logs
openclaw gateway logs --follow

# If needed, start or restart the gateway
openclaw gateway restart
```

10 Add Swap Space (Recommended for Low-RAM Plans)

If your VPS has 1–2 GB of RAM, adding swap space helps prevent crashes during intensive tasks:

```
# Create a 2 GB swap file
sudo fallocate -l 2G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile

# Make it permanent across reboots
echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab
```

SECTION 07

Security Essentials

This is the most important section of this guide. A VPS is an internet-connected server — and when your AI CEO is connected to Stripe, email, CRM platforms, and other sensitive business accounts, security is not optional. These steps are non-negotiable.

> 🚨 **CRITICAL WARNING**
>
> Do NOT simply install OpenClaw on a VPS, expose the dashboard to the internet, and start connecting sensitive business accounts. OpenClaw's own documentation explicitly warns against this. A proper security setup is required before connecting any important integrations.

🔐 Step 1 — Secure SSH Access

Your SSH connection is the door to your server. Lock it down:

- Use SSH key authentication only — disable password-based SSH login
- Protect your hosting provider account with two-factor authentication (2FA)
- Keep your private SSH key file secure on your local machine
- Consider installing **fail2ban** to block repeated failed login attempts

```
# Disable password authentication in SSH config
sudo nano /etc/ssh/sshd_config

# Find and set these values:
PasswordAuthentication no
PubkeyAuthentication yes

# Restart SSH to apply changes
sudo systemctl restart sshd
```

🔥 Step 2 — Configure the Firewall

Only allow the traffic your server actually needs. Everything else should be blocked:

```
# Enable UFW firewall and allow only SSH
sudo ufw allow ssh
sudo ufw enable

# Keep the OpenClaw Gateway port on localhost ONLY — never expose it publicly
# The Gateway listens on port 18789 by default
# Do NOT run: sudo ufw allow 18789 ← this would expose it publicly

# Check firewall status
sudo ufw status verbose
```

🔒 Step 3 — Never Expose the OpenClaw Dashboard Publicly

OpenClaw's documentation specifically warns against exposing the Gateway directly to the internet. Instead, use one of these two secure access methods:

Option A — SSH Tunnel (no additional software needed):

```
# Run this on your LOCAL computer to create a secure tunnel
ssh -N -L 18789:127.0.0.1:18789 openclaw@YOUR.SERVER.IP

# Then open your browser and visit: http://localhost:18789
# The connection is private — only accessible from your computer
```

Option B — Tailscale (recommended for ongoing access):

Tailscale is a free tool that creates a private, encrypted network between your devices. Install it on both your VPS and your computer — then access OpenClaw at a private Tailscale IP address, visible only to your devices. This is the cleanest long-term solution.

🗝️ Step 4 — Protect Your API Credentials

This is especially important for Stripe, email providers, CRM platforms, and any financial integrations.

- **Never paste API keys into Markdown files, chat messages, or command histories**
- Use **restricted API keys** with only the permissions your AI CEO actually needs (read-only where possible)
- Create separate API keys for each service — don't reuse keys across platforms

- For financial accounts, use read-only access for reporting; require separate human approval for any action that moves money
- Store credentials using environment variables or OpenClaw's secure configuration — not in plain text files

> ⚠️ **ABOUT OPENCLAW'S STATE STORAGE**
>
> OpenClaw's documentation notes that its state — which can include OAuth tokens and authentication data — is stored in a local SQLite database. This means your state directory (`~/.openclaw/`) must be treated as sensitive. Restrict its permissions and ensure backups are encrypted.

💾 Step 5 — Encrypted Backups

- Back up your `~/.openclaw/` directory regularly using your provider's snapshot tool or a manual backup
- Treat backups as sensitively as the live server — they may contain API tokens and conversation history
- Test your backups occasionally by verifying you can restore from them
- Use OpenClaw's built-in backup command: `openclaw backup create`

🔍 Step 6 — Run OpenClaw's Security Audit

```
# Run OpenClaw's built-in security audit
openclaw security audit --deep

# Review and address any warnings it surfaces
```

> 🧯 **PROMPT INJECTION RISK**
>
> An AI agent that reads external information (emails, web pages, messages) and has access to powerful tools can be vulnerable to "prompt injection" — where malicious content in an email or message attempts to trick the agent into taking unauthorized actions. OpenClaw's documentation covers this risk. Configure your agent's tool permissions carefully and require human approval for sensitive or financial actions.

Security Setup Checklist

- [ ] SSH key authentication enabled; password login disabled
- [ ] Hosting provider account protected with 2FA
- [ ] UFW firewall enabled; only SSH port open publicly
- [ ] OpenClaw Gateway NOT exposed to the public internet
- [ ] Private access configured (SSH tunnel or Tailscale)
- [ ] Non-root user created for running OpenClaw
- [ ] Restricted API keys in use for all integrations
- [ ] No API credentials stored in plain-text Markdown files
- [ ] Backups configured and encryption applied
- [ ] openclaw security audit --deep completed
- [ ] Server updates and patches kept current

SECTION 08

Mac mini vs. VPS — Which is Right for You?

Use this guide to help decide which path makes the most sense for your specific situation.

FACTOR	🖥️ MAC MINI (COURSE PATH)	☁️ VPS
Technical comfort level	Beginner to intermediate — the course guides you through every step	Intermediate to advanced — requires comfort with SSH, Linux terminal, and server administration
Upfront cost	$500–$800+ (one-time hardware purchase)	$0 upfront — monthly subscription of $5–$25
Ongoing cost	None (hardware owned outright; electricity only)	$60–$300/year ongoing
Course compatibility	Fully compatible — built for this path	Workspace/AI files transfer; installation steps are different and not covered in the course
Always-on availability	Only when physically powered on at your location	Yes — 24/7 uptime managed by data center
Physical maintenance	You manage the hardware directly	Hardware managed by hosting provider
Security complexity	Standard home network security	More involved — requires firewall, SSH hardening, private gateway access
macOS-specific features	Full macOS app support	Not available (Linux only)
CyrusHQ personal experience	Fully tested — this is what we use	Not personally tested — guidance based on official documentation and research

💡 OUR HONEST TAKE

If you're newer to the technical side of things, the Mac mini is the cleaner, more supported path — and it's what CyrusHQ stands behind from personal experience. If you're comfortable with server environments and want to avoid the upfront hardware cost, a VPS is a legitimate option. The course files and AI CEO architecture transfer directly — it's the installation and security setup where the paths diverge.

SECTION 09

A Note About the Course

The *Build Your AI CEO* course was designed and built around a Mac mini installation. Every terminal walkthrough, screenshot, and installation step was created in a macOS environment — because that's what we use, have tested, and can stand behind.

The VPS path is not covered inside the course.

The good news is that everything you learn in the course about *building* your AI CEO — writing the identity files, structuring the workspace, setting up memory, creating subagents, connecting integrations — applies exactly the same way on a VPS. The knowledge transfers. The workspace structure is identical. Only the installation and server setup steps are different.

⚠ IMPORTANT DISCLAIMER

CyrusHQ has not personally tested OpenClaw on a VPS. Our AI CEO (Cyrus) and all of our agents run on Mac mini computers. The information in this guide comes from OpenClaw's official documentation and independent research. We are confident in the quality of that research, but we cannot guarantee your specific experience or troubleshoot VPS-specific installation issues. For VPS-specific support, OpenClaw's official documentation, community forums, and your hosting provider's support team are your best resources.

SECTION 10

Next Steps & Official Resources

If you decide to proceed with a VPS:

1. Choose your hosting provider based on the comparison above
2. Follow OpenClaw's dedicated installation guide for your provider (links below)
3. Complete the security checklist in Section 7 before connecting any sensitive integrations
4. Set up your workspace using the same Markdown files from the course
5. Run `openclaw doctor` and `openclaw security audit --deep` before going live

If you decide to go with a Mac mini:

1. The course covers everything you need from start to finish
2. You'll have the benefit of working in the same environment CyrusHQ uses
3. Installation is more straightforward and fully documented in the course

Official OpenClaw Resources

Resource	URL
Main documentation	docs.openclaw.ai
Install overview	docs.openclaw.ai/install
DigitalOcean guide	docs.openclaw.ai/install/digitalocean
Hetzner guide	docs.openclaw.ai/install/hetzner
Hostinger guide	docs.openclaw.ai/install/hostinger
Security documentation	docs.openclaw.ai/security
Linux setup guide	docs.openclaw.ai/getting-started/linux

🙌 **QUESTIONS?**

> If you'd like to talk through which option is right for your specific situation, reach out to us directly. We're happy to help you make the most informed decision for your business.

CyrusHQ — Build Your AI CEO

This guide is provided for informational purposes based on OpenClaw's official documentation and independent research.
CyrusHQ has not personally tested an OpenClaw VPS installation and cannot guarantee specific outcomes.
Always refer to docs.openclaw.ai for the most current installation instructions.